SQL - Day - 12 Hands On - Sana Naveen

Problem 1

Problem Statement :

Create Database **EventDb** and create tables as per schema given below.

Requirements

1.Add entity class **UserInfo** and add public properties:

Property	**Type**	**Constraint**
EmailId	varchar	**PK**
UserName	varchar	Not Null, Length: 1 to 50 char.
Role	varchar	Not Null, (Admin,Participant)-CHECK
Password	varchar	Not Null Length: 6 to 20 char

2. Add entity class **EventDetails** and add below public properties.

Property Name	**Type**	**Constraint**
EventId	int	**PK**
EventName	varchar	Not Null, Length: 1 to 50 char.
EventCategory	varchar	Not Null, Length: 1 to 50 char.
EventDate	datetime	Not Null
Description	varchar	Nullable
Status	varchar	Active or In-Active, CHECK

4. Add entity class **SpeakersDetails** and add below public properties.

Property Name	**Type**	**Constraints**
SpeakerId	int	**PK**
SpeakerName	varchar	Not Null, Length: 1 to 50 char.

5. Add entity class **SessionInfo** and add below public properties.

Property Name	**Type**	**Constraints**

SessionId	int	**PK**
EventId	int	Not Null **(FK)**
SessionTitle	varchar	Not Null, Length: 1 to 50 char.
SpeakerId	int	Not Null **(FK)**
Description	varchar	Nullable
SessionStart	datetime	Not Null (Time Slot)
SessionEnd	datetime	Not Null (Time Slot)
SessionUrl	varchar	

6. Add entity class **ParticipantEventDetails** and add below public properties.

Property Name	**Type**	**Constraints**
Id	int	**(PK)**
ParticipantEmailId	varchar	Not Null **(FK)**
EventId	int	Not Null **(FK)**
SessionId	Int	Not Null **(FK)**
IsAttended	bit	Yes or No, CHECK

🛠️ Technical Constraints

- Use SQL Server.
- Use appropriate data types (INT, VARCHAR, DATE, DECIMAL).
- Follow normalization up to 2NF.
- Use ANSI SQL syntax only.

Expectations:

- Proper table structure with relationships.
- Valid constraints implementation.
- Correct data insertion and retrieval queries.

Learning Outcome

You will understand database structure, basic normalization, constraints, and ANSI SQL queries.

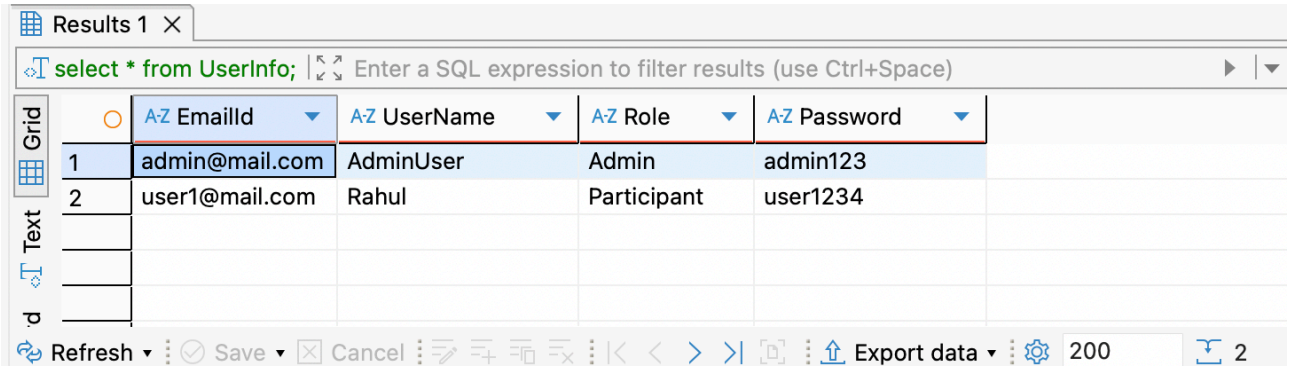
Code :

```
CREATE DATABASE EventDb;
GO

USE EventDb;
GO

CREATE TABLE UserInfo (
   EmailId VARCHAR(100) PRIMARY KEY,

   UserName VARCHAR(50) NOT NULL
      CHECK (LEN(UserName) BETWEEN 1 AND 50),

   Role VARCHAR(20) NOT NULL
      CHECK (Role IN ('Admin','Participant')),

   Password VARCHAR(20) NOT NULL
      CHECK (LEN(Password) BETWEEN 6 AND 20)
);

CREATE TABLE EventDetails (
   EventId INT PRIMARY KEY,

   EventName VARCHAR(50) NOT NULL
      CHECK (LEN(EventName) BETWEEN 1 AND 50),

   EventCategory VARCHAR(50) NOT NULL
      CHECK (LEN(EventCategory) BETWEEN 1 AND 50),

   EventDate DATETIME NOT NULL,

   Description VARCHAR(255),

   Status VARCHAR(20)
      CHECK (Status IN ('Active','In-Active'))
);
```

```sql
CREATE TABLE SpeakersDetails (
   SpeakerId INT PRIMARY KEY,

   SpeakerName VARCHAR(50) NOT NULL
      CHECK (LEN(SpeakerName) BETWEEN 1 AND 50)
);

CREATE TABLE SessionInfo (
   SessionId INT PRIMARY KEY,

   EventId INT NOT NULL,

   SessionTitle VARCHAR(50) NOT NULL
      CHECK (LEN(SessionTitle) BETWEEN 1 AND 50),

   SpeakerId INT NOT NULL,

   Description VARCHAR(255),

   SessionStart DATETIME NOT NULL,

   SessionEnd DATETIME NOT NULL,

   SessionUrl VARCHAR(255),

   FOREIGN KEY (EventId)
      REFERENCES EventDetails(EventId),

   FOREIGN KEY (SpeakerId)
      REFERENCES SpeakersDetails(SpeakerId)
);

CREATE TABLE ParticipantEventDetails (
   Id INT PRIMARY KEY,

   ParticipantEmailId VARCHAR(100) NOT NULL,

   EventId INT NOT NULL,

   SessionId INT NOT NULL,

   IsAttended BIT
      CHECK (IsAttended IN (0,1)),

   FOREIGN KEY (ParticipantEmailId)
      REFERENCES UserInfo(EmailId),

   FOREIGN KEY (EventId)
      REFERENCES EventDetails(EventId),

   FOREIGN KEY (SessionId)
      REFERENCES SessionInfo(SessionId)
);

INSERT INTO UserInfo VALUES
('admin@mail.com','AdminUser','Admin','admin123'),
('user1@mail.com','Rahul','Participant','user1234');

INSERT INTO EventDetails VALUES
```

```
(1,'Tech Conference','Technology','2026-07-10','Annual Tech Event','Active'),
(2,'AI Summit','Artificial Intelligence','2026-08-15','AI Innovations','Active');

INSERT INTO SpeakersDetails VALUES
(1,'Dr. John Smith'),
(2,'Dr. Sarah Lee');

INSERT INTO SessionInfo VALUES
(1,1,'Future of AI',1,'AI trends',
 '2026-07-10 10:00','2026-07-10 11:00','www.session1.com'),

(2,2,'Machine Learning Basics',2,'ML introduction',
 '2026-08-15 09:00','2026-08-15 10:00','www.session2.com');

INSERT INTO ParticipantEventDetails VALUES
(1,'user1@mail.com',1,1,1),
(2,'user1@mail.com',2,2,0);

select * from UserInfo;
```

Output Screenshot :



Explanation :

The EventDb database is designed to manage event-related information using multiple related tables. The UserInfo table stores user details such as email, username, role, and password. EventDetails contains information about events, while SpeakersDetails stores speaker information. The SessionInfo table links events and speakers to specific sessions using foreign keys. ParticipantEventDetails records participant registrations and attendance for sessions. Primary keys ensure unique records, foreign keys maintain relationships between tables, and CHECK constraints enforce valid data values, creating a structured and normalized relational database.